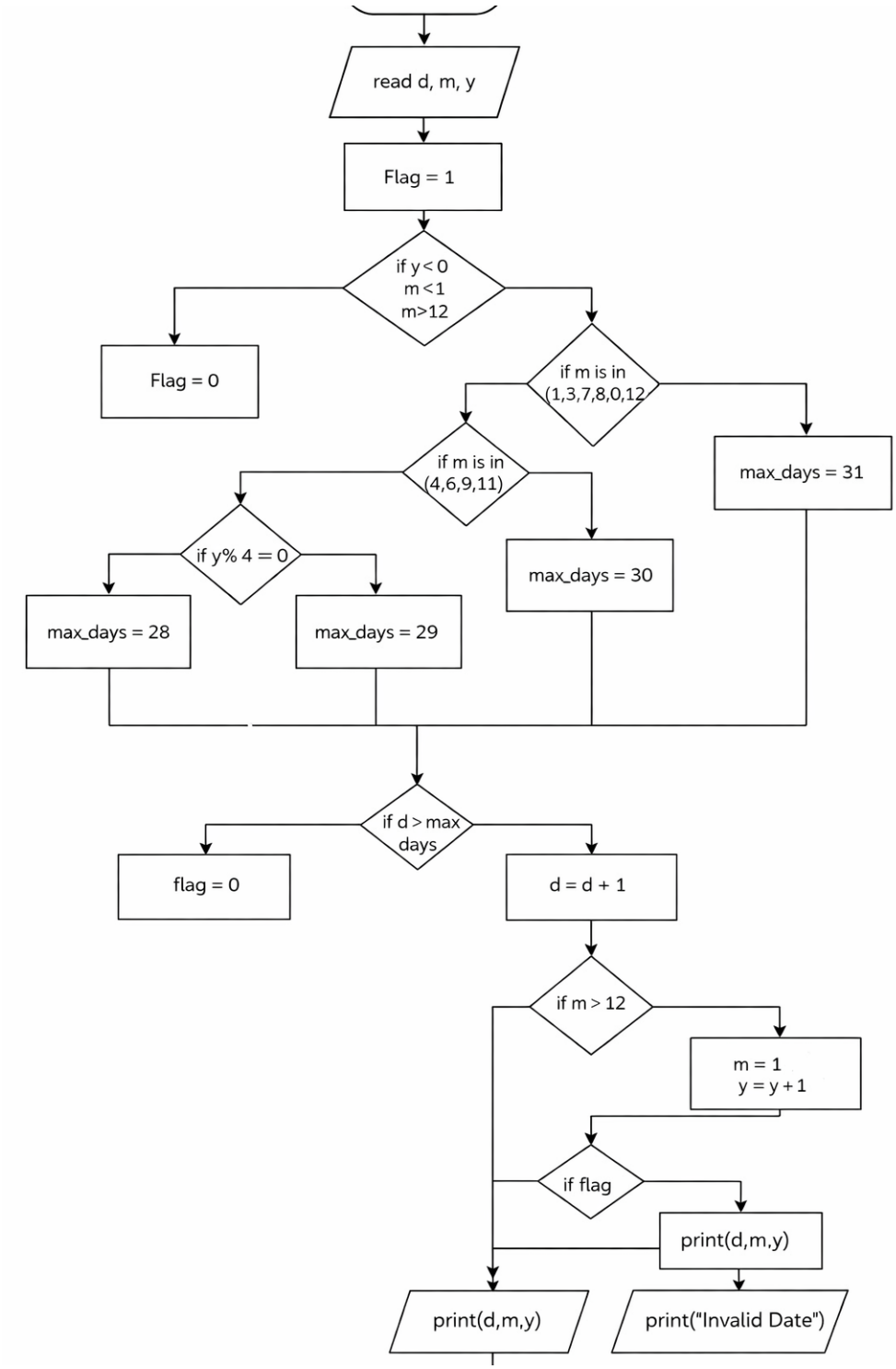
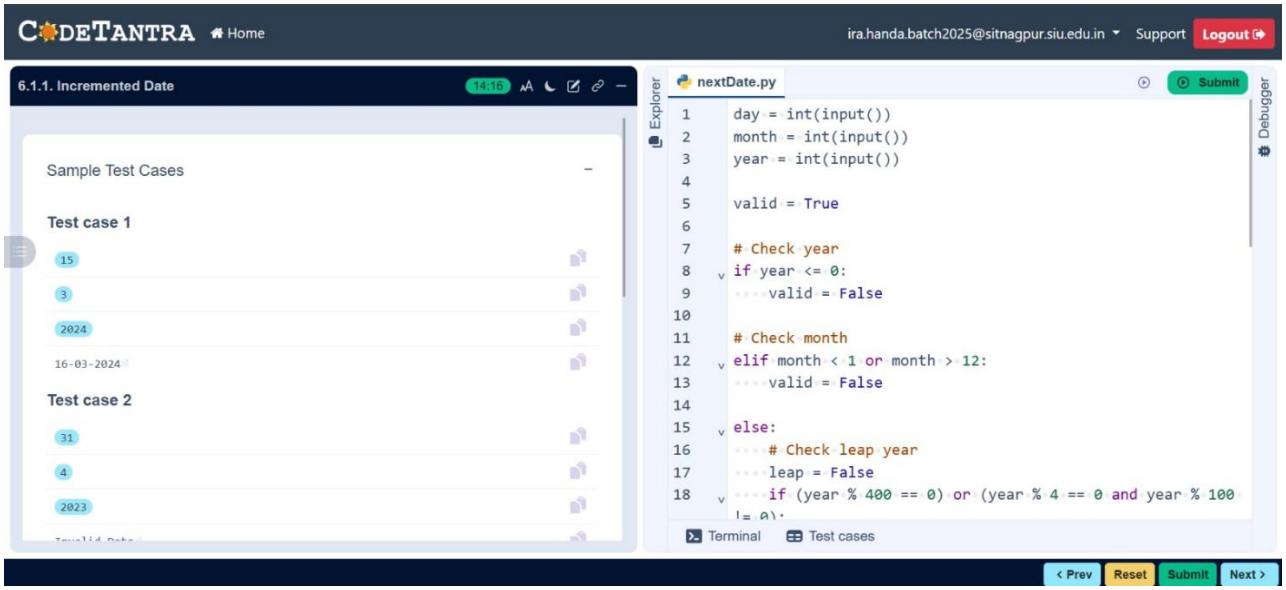
EXPERIMENT 6.1.1

FLOWCHART:



CODETANTRA:



ALGORITHM:

Incremented Date

Step 1: Start

Step 2: Read d (day), m (month), y (year)

Step 3: Set flag = 1

Step 4: Check year validity

- If y <= 0
    - Set flag = 0

Step 5: Check month validity

- If m < 1 or m > 12
    - Set flag = 0

Step 6: Determine maximum days in the month

- If m is in {1, 3, 5, 7, 8, 10, 12}
    - max_days = 31
- Else if m is in {4, 6, 9, 11}
    - max_days = 30
- Else if m == 2
    - If (y % 400 == 0) or (y % 4 == 0 and y % 100 != 0)

        - max_days = 29 *(leap year)*
    - Else
        - max_days = 28

Step 7: Validate the day

- If d > max_days or d < 1
    - Set flag = 0

Step 8: If flag == 1 (date is valid)

- Set d = d + 1
- If d > max_days
    - d = 1
    - m = m + 1
- If m > 12
    - m = 1
    - y = y + 1
- Print d-m-y

Step 9: Else

- Print "Invalid Date"

Step 10: Stop

EXPERIMENT 6.2.1

ALGORITHM: Factorial of a Number

Step 1: Start

Step 2: Read an integer n

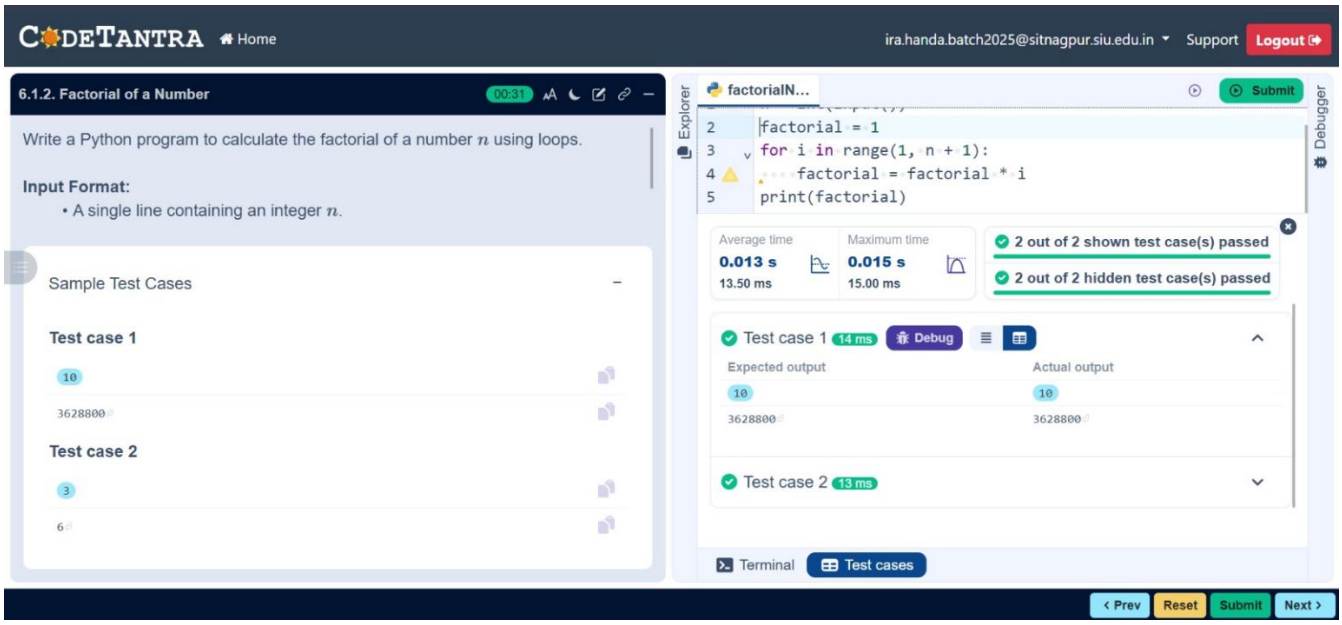
Step 3: Set factorial = 1

Step 4: Repeat the following steps for i from 1 to n, factorial = factorial × i

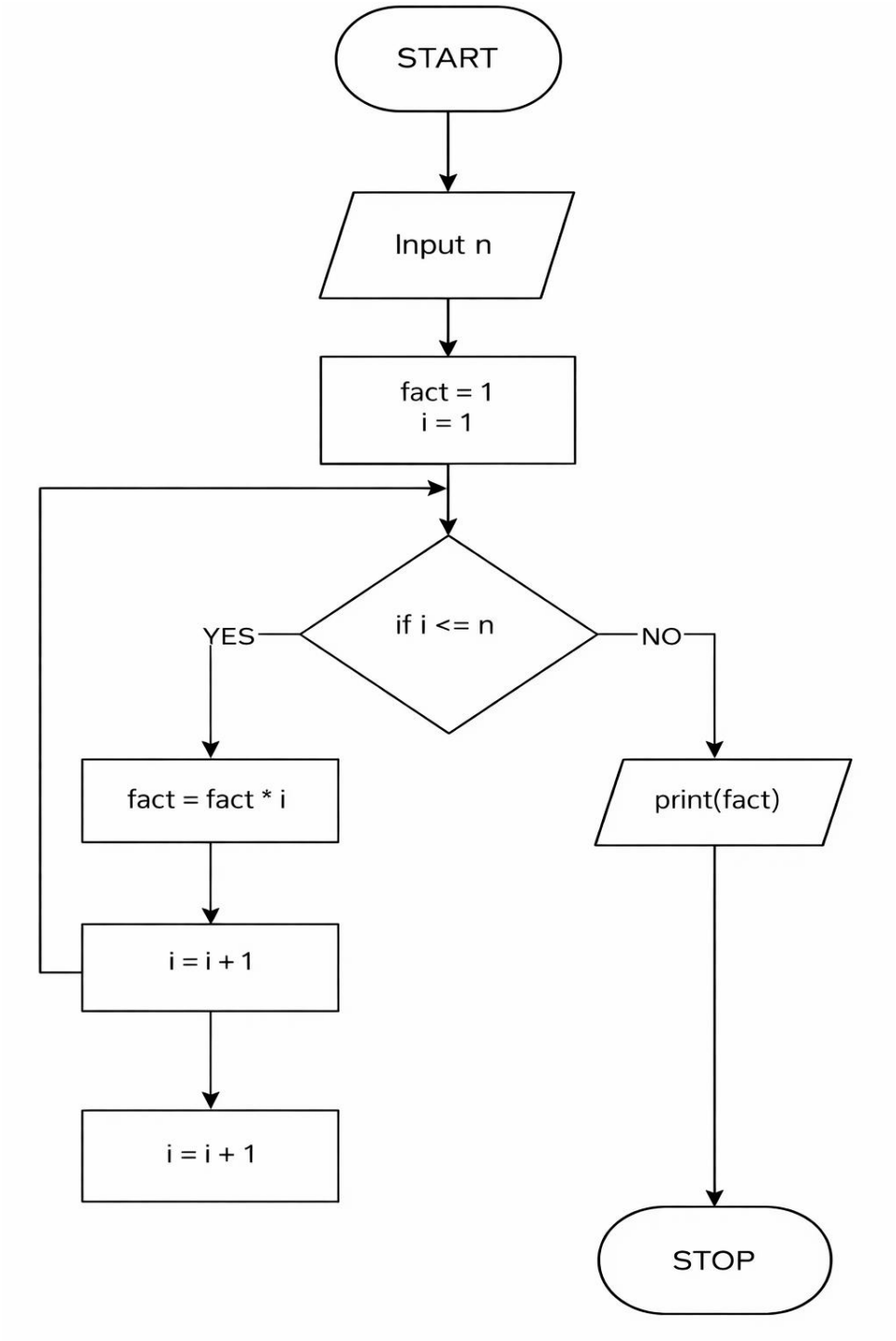
Step 5: Print factorial

Step 6: Stop

CODETANTRA:


CODETANTRA
Home
ira.handa.batch2025@sitnagpur.siu.edu.in
Support
Logout
6.1.2. Factorial of a Number
00:31
Write a Python program to calculate the factorial of a number n using loops.
Input Format:
• A single line containing an integer n.
Sample Test Cases
Test case 1
10
3628800
Test case 2
3
6
factorialN...
Submit
Explorer
Debugger
factorial = 1
for i in range(1, n + 1):
factorial = factorial * i
print(factorial)
Average time
0.013 s
13.50 ms
Maximum time
0.015 s
15.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1 14 ms
Debug
Expected output
Actual output
10
3628800
10
3628800
Test case 2 13 ms
Terminal
Test cases
Prev
Reset
Submit
Next

FLOWCHART:


START
Input n
fact = 1
i = 1
if i <= n
YES
NO
fact = fact * i
i = i + 1
i = i + 1
print(fact)
STOP